<u>Recall:</u> $1 \le p < n$, $p^* = \dfrac{np}{n-p}$

$U \subset \mathbb{R}^n$ bdd open set with $\partial U \in C^1$

If $u \in W^{1,p}(U)$, then $u \in L^{p^*}(U)$ and

$$\|u\|_{L^{p^*}(U)} \le C \|u\|_{W^{1,p}(U)} \quad \text{(Sobolev inequality)}$$

for some $C > 0$. So $W^{1,p}(U) \hookrightarrow L^{p^*}(U)$.

<u>Def:</u> If $X \subset Y$ are normed linear spaces, we say that $X$ is continuously embedded in $Y$ if the inclusion $i: X \to Y$ is a bdd linear map, i.e.,

$$\|u\|_Y \leq C\|u\|_X \quad \forall u \in X$$

for some constant $C > 0$. Then we write $X \hookrightarrow Y$.

# Morrey's Inequality

**Thm:** Let $n < p \leq \infty$ and set $\gamma := 1 - \frac{n}{p} \in (0,1)$. Then $\exists$ a constant $C > 0$ st

$$\|u\|_{C^{0,\gamma}(\mathbb{R}^n)} \leq C \|u\|_{W^{1,p}(\mathbb{R}^n)}$$

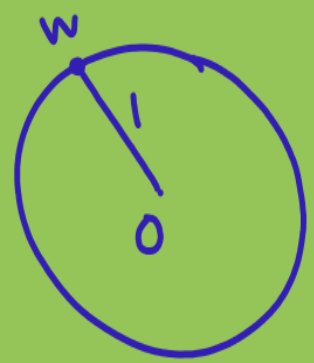
for all $u \in C^1(\mathbb{R}^n)$.

**Pf:** 1). First we fix $x \in \mathbb{R}^n$ and $r > 0$,

and estimate $\displaystyle\int_{B_r(x)} |u(y) - u(x)|\, dx.$

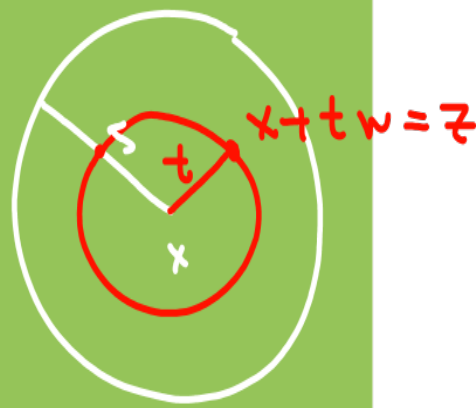
$$|u(y) - u(x)| = |u(x+sw) - u(x)|$$

$$= \left| \int_0^s \underbrace{\frac{d}{dt} u(x+tw)}_{Du(x+tw)\cdot w} dt \right|$$




$y = x + sw$

$$\leq \int_0^s \underbrace{|Du(x+tw)\cdot w|}_{\leq |Du(x+tw)|\underbrace{|w|}_{1} = |Du(x+tw)|} \, dt$$




$$|u(x+sw)-u(x)| \leq \int_0^s |Du(x+tw)|\,dt$$

$$\int_{\partial B_1(0)} |u(x+sw)-u(x)|\,dS(w) \leq \int_{\partial B_1(0)} \int_0^s |Du(x+tw)|\,dt\,dS(w) = \int_0^s \int_{\partial B_1(0)} |Du(x+tw)|\,dS(w)\,dt$$

$$= \int_0^s \int_{\partial B_1(0)} |Du(\underbrace{x+tw}_{z})| \frac{n\omega_n t^{n-1}}{n\omega_n \underbrace{t^{n-1}}_{|z-x|}} \, dS(w)\, dt$$

$z = x + tw$

$tw = z - x$

$t = |z-x| \quad (|w|=1)$

$$= \frac{1}{n\omega_n} \int_0^s \int_{\partial B_1(0)} \frac{|Du(z)|}{|z-x|^{n-1}} \, n\omega_n t^{n-1} \, dS(w)\, dt$$

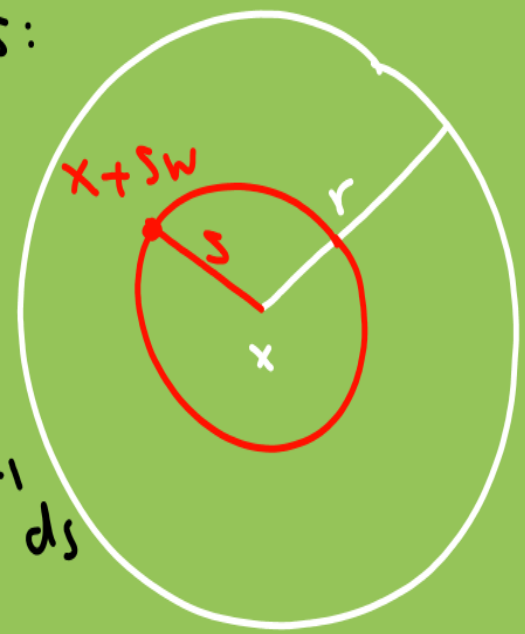
$$= \frac{1}{n\omega_n} \int_{B_s(x)} \frac{|Du(z)|}{|z-x|^{n-1}} \, dz \leq \frac{1}{n\omega_n} \int_{B_r(x)} \frac{|Du(z)|}{|z-x|^{n-1}} \, dz$$

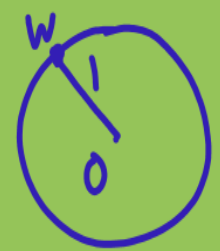
$$\int_{\partial B_1(0)} |u(x+sw)-u(x)|\,dS(w) \le \frac{1}{n\omega_n}\int_{B_r(x)} \frac{|Du(z)|}{|z-x|^{n-1}}\,dz$$

Multiply by $n\omega_n s^{n-1}$ and integrate w.r.t. $s$:

$$\int_0^r \int_{\partial B_1(0)} |u(\overbrace{x+sw}^{y})-u(x)|\, n\omega_n s^{n-1}\,dS(w)\,ds \le \int_{B_r(x)} \frac{|Du(z)|}{|z-x|^{n-1}}\,dz \underbrace{\int_0^r s^{n-1}\,ds}_{r^n}$$

$$\int_{B_r(x)} |u(y)-u(x)|\,dy \le r^n \int_{B_r(x)} \frac{|Du(z)|}{|z-x|^{n-1}}\,dz$$

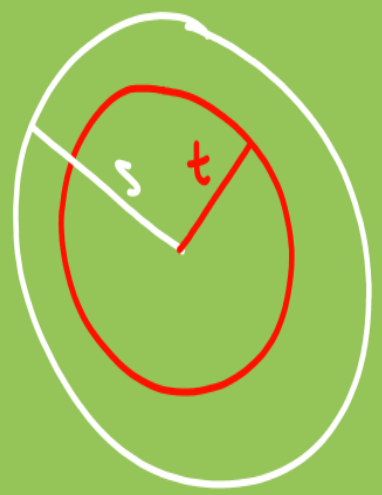


$$\|u\|_{C^{0,\gamma}(\mathbb{R}^n)} = \sup_{x\in\mathbb{R}^n} |u(x)| + \underbrace{\sup_{\substack{x,y\in\mathbb{R}^n \\ x\neq y}} \frac{|u(x)-u(y)|}{|x-y|^\gamma}}_{[u]_{C^{0,\gamma}(\mathbb{R}^n)}}$$

$$\int_{B_r(0)} f(|x|)\,dx = \int_0^s \int_{\partial B_s(0)} f(t)\,dS\,dt$$

$$= \int_0^s f(t)\,\underbrace{n\omega_n t^{n-1}}_{\text{area of } \partial B_t}\,dt$$




$\omega_n$ = volume of $B_1(0)$

2). Next we estimate $\sup_{x \in \mathbb{R}^n} |u(x)|$.

$$|u(x)| = |u(x) - u(y) + u(y)| \leq |u(x) - u(y)| + |u(y)|$$

Integrate over $B_r(x)$:

$$\int_{B_r(x)} |u(x)|\, dy \leq \int_{B_r(x)} |u(x) - u(y)|\, d + \int_{B_r(x)} |u(y)|\, dy$$

$$\omega_n r^n |u(x)| \leq r^n \int_{B_r(x)} \frac{|Du(z)|}{|z - x|^{n-1}}\, dz + \int_{B_r(x)} |u(y)|\, dy$$

$$|u(x)| \leq \frac{1}{\omega_n} \int_{B_r(x)} \frac{|Du(z)|}{|z-x|^{n-1}}\, dz + \frac{1}{\omega_n r^n} \int_{B_r(x)} |u(y)|\, dy$$

$$\int_{B_r(x)} \frac{|Du(z)|}{|z-x|^{n-1}}\, dz$$

$$\leq \left( \int_{B_r(x)} |Du|^p\, dz \right)^{1/p} \left( \int_{B_r(x)} \frac{dz}{|z-x|^{(n-1)p/(p-1)}} \right)^{(p-1)/p}$$

$$\frac{1}{p} + \frac{1}{q} = 1 \Rightarrow q = \frac{p}{p-1}$$

$$\|u\|^p_{W^{1,p}(\mathbb{R}^n)} = \int_{\mathbb{R}^n} |u|^p\, dx + \int_{\mathbb{R}^n} |Du|^p\, dx$$

$$\int_{B_r(x)} \frac{dz}{(z-x)^{(n-1)p/(p-1)}} = \int_0^r \int_{\partial B_r(x)} \frac{dS}{s^{(n-1)p/(p-1)}}\, ds = \int_0^r \frac{n\omega_n s^{n-1}}{s^{(n-1)p/(p-1)}}\, ds$$

$$= n\omega_n \int_0^r s^{n-(n-1)p/(p-1)-1}\, ds = n\omega_n \frac{s^{n-(n-1)p/(p-1)}}{n-(n-1)p/(p-1)} \Bigg|_0^r$$

$$= C r^{n-(n-1)p/(p-1)}$$

$$\frac{(n-1)p}{p-1} \overset{?}{<} n$$

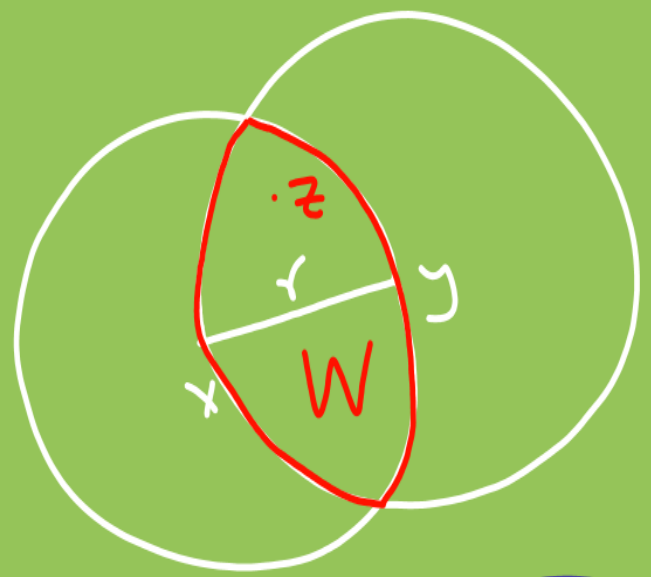
$$np-p \overset{?}{<} np-n$$

$$p \overset{?}{>} n \checkmark$$

$$\int_{B_r(x)} |u(y)|\, dy \leq \left( \int_{B_r(x)} |u(y)|^p dy \right)^{1/p} \underbrace{\left( \int_{B_r(x)} dy \right)^{(p-1)/p}}_{(\omega_n r^n)^{(p-1)/p}}$$

So $\sup_{x \in \mathbb{R}^n} |u(x)| \leq C \|u\|_{W^{1,p}(\mathbb{R}^n)}$

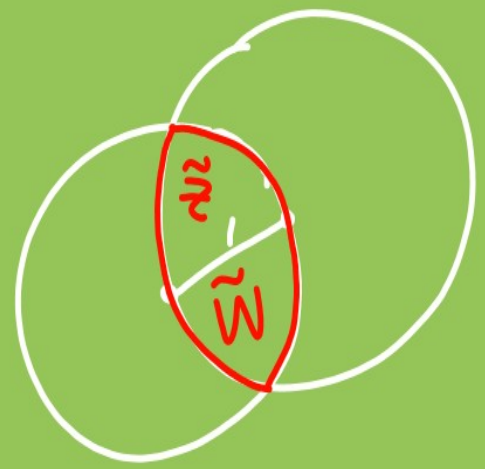
3). Finally we estimate $[u]_{C^{0,\gamma}(\mathbb{R}^n)} := \sup_{\substack{x,y \in \mathbb{R}^n \\ x \neq y}} \frac{|u(x)-u(y)|}{|x-y|^\gamma}$.




$$\int_{B_r(x)} |u(z)-u(x)|\,dz \leq r^n \int_{B_r(x)} \frac{|Du(z)|}{|z-x|^{n-1}}\,dz$$

$$\underbrace{\int_W |u(x)-u(y)|\,dz}_{|u(x)-u(y)|\,\mathrm{vol}(W) = Cr^n|u(x)-u(y)|} \leq \int_W |u(x)-u(z)|\,dz + \int_W |u(z)-u(y)|\,dz$$

$$\mathrm{vol}(W) = \int_W dz = \int_{\tilde{W}} r^n \, d\tilde{z} = r^n \underbrace{\mathrm{vol}(\tilde{W})}_{\text{constant}}$$




$$z = r\tilde{z}, \quad dz = r^n \, d\tilde{z}$$

$$= C r^n$$

$$C r^n |u(x)-u(y)| \leq \int_{B_r(x)} |u(x)-u(z)|\,dz + \int_{B_r(x)} |u(y)-u(z)|\,dz$$

$$\leq r^n \underbrace{\int_{B_r(x)} \frac{|Du(z)|}{|z-x|^{n-1}}\,dz}_{\leq C r^{[n-(n-1)p/(p-1)](p-1)/p} \|u\|_{W^{1,p}(\mathbb{R}^n)}} + r^n \int_{B_r(y)} \frac{|Du(z)|}{|z-y|^{n-1}}\,dz$$

$$|u(x)-u(y)| \leq C r^{\gamma} \|u\|_{W^{1,p}(\mathbb{R}^n)} = C|x-y|^{\gamma} \|u\|_{W^{1,p}(\mathbb{R}^n)}$$

$$n(p-1)/p - (n-1) = 1 - n/p = \gamma$$

$$\frac{|u(x)-u(y)|}{|x-y|^{\gamma}} \leq C\|u\|_{W^{1,p}(\mathbb{R}^n)}$$

Taking sup over all $x \neq y$ in $\mathbb{R}^n$ gives $[u]_{C^{0,\gamma}(\mathbb{R}^n)} \leq C\|u\|_{W^{1,p}(U)}$.

4. Combining (2) and (3) gives

$$\|u\|_{C^{0,\gamma}(\mathbb{R}^n)} = \sup_{x\in\mathbb{R}^n} |u(x)| + [u]_{C^{0,\gamma}(\mathbb{R}^n)} \leq C\|u\|_{W^{1,p}(\mathbb{R}^n)} \quad \#$$

**Thm:** Let $U \subset \mathbb{R}^n$ be a bdd open set with $\partial U \in C^1$.

Let $n < p \leq \infty$ and set $\gamma := 1 - n/p \in (0,1)$. Then $\exists$ a constant $C > 0$ and for each $u \in W^{1,p}(U)$ a function $\tilde{u} \in W^{1,p}(U)$ st

1). $\tilde{u} = u$ a.e. in $U$,

2). $\|\tilde{u}\|_{C^{0,\gamma}(\bar{U})} \leq C \|u\|_{W^{1,p}(U)}$.

<u>Pf:</u> Since $\partial U \in C^1$, $\exists$ an extension $\bar{u}$ of $u$ to $\mathbb{R}^n$ st

1). $\bar{u} = u$ a.e. in $U$,

2). $\bar{u}$ has compact support in $\mathbb{R}^n$,

3). $\|\bar{u}\|_{W^{1,p}(\mathbb{R}^n)} \leq C \|u\|_{W^{1,p}(U)}$ for some constant $C > 0$.

Since $\bar{u}$ has compact support, $\exists$ a seq $(u_j)$ in $C^1(\mathbb{R}^n)$ st

$u_j \to \bar{u}$ in $W^{1,p}(\mathbb{R}^n)$. By Morrey's inequality,

$$\underbrace{\|u_j\|_{C^{0,\gamma}(\mathbb{R}^n)}}_{\to \|\tilde{u}\|_{C^{0,\gamma}(\mathbb{R}^n)}} \leq C \underbrace{\|u_j\|_{W^{1,p}(\mathbb{R}^n)}}_{\to \|\tilde{u}\|_{W^{1,p}(\mathbb{R}^n)}} \qquad \text{——— (1)}$$

$$\|u_j - u_k\|_{C^{0,\gamma}(\mathbb{R}^n)} \leq C \|u_j - u_k\|_{W^{1,p}(\mathbb{R}^n)} \qquad \text{——— (2)}$$

for some constant $C > 0$. Since $u_j$ converges in $W^{1,p}(\mathbb{R}^n)$,

$\|u_j - u_k\|_{W^{1,p}(\mathbb{R}^n)} \to 0$ as $j, k \to \infty$. Then (2) shows that

$(u_j)$ is a Cauchy seq in $C^{0,\gamma}(\mathbb{R}^n)$. Since $C^{0,\gamma}(\mathbb{R}^n)$ is a Banach sp,

then $u_j \to \tilde{u}$ in $C^{0,\gamma}(\mathbb{R}^n)$ for some $\tilde{u} \in C^{0,\gamma}(\mathbb{R}^n)$. Now letting $j \to \infty$ gives

$$\|\tilde{u}\|_{C^{0,\gamma}(\mathbb{R}^n)} \leq C\|\tilde{u}\|_{W^{1,p}(\mathbb{R}^n)} \leq C\|u\|_{W^{1,p}(U)}$$

It remains to show that $\tilde{u} \in W^{1,p}(U)$ and $\tilde{u} = u$ a.e. in $U$.

We have $u_j \to \tilde{u}$ in $C^{0,\gamma}(\mathbb{R}^n)$ and $u_j \to \bar{u}$ in $W^{1,p}(\mathbb{R}^n)$,

in particular, $u_j(x) \to \tilde{u}(x)$ for all $x \in \mathbb{R}^n$ and $u_j(x) \to \bar{u}(x)$ for a.a. $x \in \mathbb{R}^n$ for a renamed subseq. So $\tilde{u} = \bar{u}$ a.e. in $\mathbb{R}^n$. Since $\bar{u} = u$ a.e. in $U$, then $\tilde{u} = u$ a.e. in $U$.

and $\tilde{u} \in W^{1,p}(U)$. #